
Flux & Key: Reference-Guided Image Composition with K/V Injection and Domain Guidance

Hamna Aqeel*

School of Science and Engineering
Lahore University of Management Sciences
24280028@lums.edu.pk

Rabia Aslam*

School of Science and Engineering
Lahore University of Management Sciences
24280019@lums.edu.pk

Syeda Attqa Batool*

School of Science and Engineering
Lahore University of Management Sciences
24280050@lums.edu.pk

Amin Ahmad Qasmi*

School of Science and Engineering
Lahore University of Management Sciences
24280005@lums.edu.pk

Abstract

Reference-guided image composition requires inserting a specific foreground object into a target scene while preserving object identity, visual harmonization, and background consistency. Existing training-free editing pipelines often prioritize either efficient structure-preserving editing or reference-guided composition, but reference appearance is commonly carried through latent initialization or prompt conditioning rather than being directly exposed to the denoising attention mechanism. We explore a lightweight extension of EEdit for FLUX that retains its RF-Inversion and caching backbone while adding reference key-value attention injection, domain-aware prompt augmentation, and tail classifier-free guidance. A single forward pass over the reference object extracts key-value memories from early FLUX joint transformer blocks, which are appended to the attention memory during denoising. We evaluate against unmodified EEdit on the TF-ICON test benchmark across Real→Cartoon, Real→Painting, Real→Sketch, and Real→Real composition settings. Results show consistent CLIP-alignment improvements, with the strongest gains in Real→Sketch and Real→Real, while EEdit remains stronger in strict background preservation for some highly stylized domains. These findings suggest that reference attention injection offers a controllable trade-off between object-level alignment and conservative background reconstruction.

1 Introduction

The ability to edit and compose images with precise, localized control has long been a central challenge in computer vision and computational photography (Meng et al., 2021). Traditional compositing tools offer pixel-level control, but require substantial manual effort and domain expertise. Diffusion-based generative models (Ho et al., 2020; Song et al., 2020; Rombach et al., 2022) have significantly lowered this barrier by enabling natural-language image editing (Brooks et al., 2023; Kawar et al., 2023). However, text-driven editing remains limited when the goal is to preserve the identity of a specific object. A prompt such as “insert sunglasses” can synthesize a plausible category instance, but it cannot reliably reproduce the exact frame geometry, color, texture, or material properties of a particular pair. This gap between *category-level* generation and *instance-level* fidelity motivates our work.

*All authors contributed equally to this work.

This limitation is especially important in reference-guided image composition, where a specific foreground object must be inserted into a target scene while preserving its identity. Such a setting arises in e-commerce visualization, personalized design, and virtual try-on applications (Islam et al., 2024), where small object-specific details can determine whether the generated composite is useful. At the same time, the object must be harmonized with the target background in terms of style, lighting, and visual domain, while non-edited regions remain stable.

Recent diffusion transformer models such as FLUX (Black Forest Labs, 2024) provide a strong foundation for this problem because they process images as latent token sequences and jointly model image and text streams through attention. This architecture improves prompt adherence and creates natural opportunities for token-level intervention. However, local reference-guided composition remains challenging: reference appearance can weaken during denoising, and modifying foreground tokens can still influence surrounding regions through global attention.

In this work, we explore a lightweight extension of EEdit (Yan et al., 2025) for instance-level reference-guided image composition. We retain EEdit as the computational backbone and introduce three inference-time modifications: reference key-value attention injection, domain-aware prompt augmentation, and tail classifier-free guidance. **A single forward pass over the reference object extracts key-value memories from early FLUX joint transformer blocks, which are then appended to the attention memory during denoising. This allows generated image tokens to attend directly to reference appearance cues while preserving the efficient RF-Inversion and caching structure of EEdit.**

We evaluate our method on the TF-ICON test benchmark (Lu et al., 2023), which provides foreground objects, masks, and target backgrounds across multiple visual domains. Our goal is not to replace EEdit as a general-purpose editing framework, **but to study whether explicit reference attention and domain-aware guidance alter its composition trade-off.** Results show improved CLIP alignment across all evaluated domains, with the strongest gains in Real→Sketch and Real→Real settings, while EEdit remains stronger in strict background preservation for some highly stylized domains.

2 Related Work

Diffusion-based image editing. Diffusion models have become the dominant framework for high-fidelity image synthesis and editing (Ho et al., 2020; Song et al., 2020; Rombach et al., 2022). Early training-free editing methods were largely developed around U-Net diffusion backbones (Ronneberger et al., 2015), where inversion and attention manipulation provided useful handles for localized semantic changes. Prompt-to-Prompt (Hertz et al., 2022) demonstrated that cross-attention maps can preserve layout during prompt edits, while Plug-and-Play Diffusion (Tumanyan et al., 2023) and MasaCtrl (Cao et al., 2023) further explored feature and attention reuse for structure-preserving generation. Benchmarks such as PIE-Bench (UB-CVML-Group, 2024) have been widely used to evaluate prompt-driven image editing in terms of background preservation and text alignment. However, PIE-Bench primarily targets single-image prompt edits, whereas our setting requires reference-guided object composition across target visual domains.

Diffusion transformers and efficient inversion-based editing. More recent models such as FLUX (Black Forest Labs, 2024) use multi-modal diffusion transformer architectures, where image and text tokens are processed jointly. This improves prompt adherence but makes editing more dependent on how information is injected into transformer attention blocks. A major limitation of inversion-based editing is its computational cost: the input image must first be mapped to a noisy latent trajectory and then denoised under the target condition. KV-Edit (Zhu et al., 2025) focuses on strict background preservation by caching and freezing background key-value representations during denoising. EEdit (Yan et al., 2025) instead studies redundancy in inversion-based editing and introduces Spatial Locality Caching, Token Index Preprocessing, and Inversion Step Skipping to reduce both spatial and temporal computation. These methods show that many tokens and timesteps are redundant during local editing. Our work adopts EEdit as the computational backbone rather than proposing a new acceleration method.

Reference-guided composition with EEdit. Reference-guided composition differs from prompt-only editing because the target object is specified by an image rather than by text alone. In this setting, the model must preserve the identity of the provided foreground object while adapting it to the target scene. EEdit already supports reference-guided composition as one of its editing scenarios, using RF-Inversion, localized recomputation, and background anchoring to efficiently preserve the target

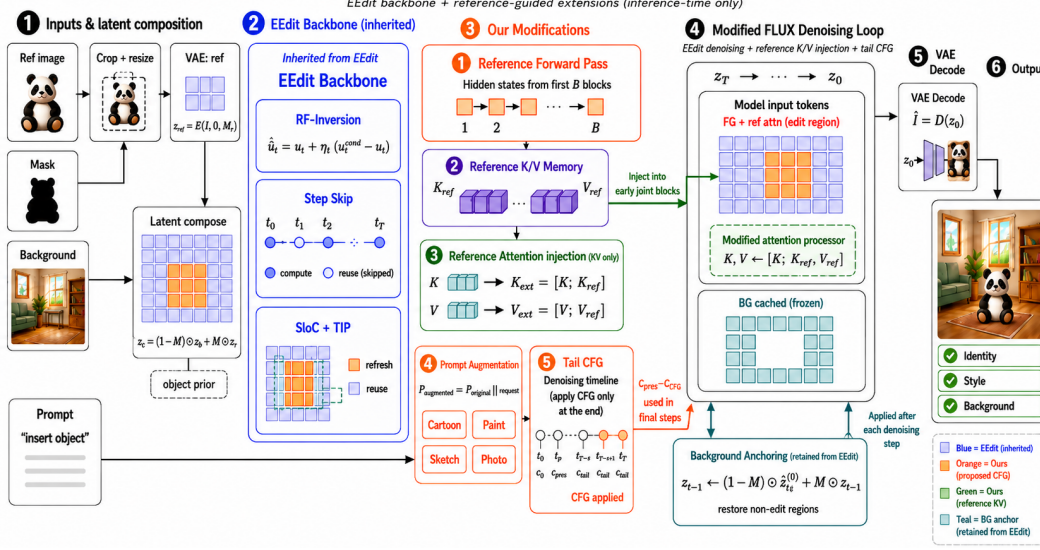


Figure 1: Overview of the proposed reference-guided image composition pipeline. Blue modules denote the inherited EEdit backbone, including RF-Inversion, inversion step skipping, and Spatial Locality Caching; orange modules denote our added prompt and guidance components; green denotes reference K/V attention injection; and teal denotes background anchoring. The figure highlights that our additions modify the EEdit denoising process directly: reference K/V memories are injected into early joint FLUX attention blocks, tail CFG guides the final denoising steps, and background anchoring preserves non-edited regions throughout generation.

scene during editing. However, in the base EEdit pipeline, reference appearance is primarily carried through the initial composite latent and prompt conditioning. During denoising, the reference object is not explicitly exposed to the transformer’s internal attention memory.

Our work is positioned as a lightweight extension of EEdit for reference-guided composition. We retain EEdit’s RF-Inversion, Spatial Locality Caching, Token Index Preprocessing, and background anchoring, but modify the denoising process by injecting reference-derived key-value memories into early FLUX joint attention blocks. We further add domain-aware prompt augmentation and tail classifier-free guidance to improve cross-domain harmonization. Unlike methods that seek to replace EEdit or establish a new general-purpose editing backbone, our goal is to explore a different operating point: preserving the efficiency of EEdit while giving the denoising process more direct access to reference-object appearance.

3 Methodology

We adopt EEdit as the underlying editing framework and extend it for instance-level image composition. Given a target background image I_b , a reference foreground image I_r , a foreground mask M_r , and a textual prompt p , our objective is to insert the reference object into the target scene while preserving both object identity and scene consistency.

EEdit provides an efficient editing backbone based on RF-Inversion, inversion step skipping, and Spatial Locality Caching (SLoC). While these components enable computationally efficient editing, they do not explicitly preserve reference-specific appearance information during denoising. As a result, object identity may gradually drift toward a generic category-level representation, particularly for objects with distinctive textures, colors, or stylistic attributes.

To address this limitation, we introduce three modifications tailored for reference-guided composition: (i) reference attention injection, which treats the reference image as an external attention memory throughout denoising, (ii) domain-aware prompt conditioning for improved cross-domain harmonization, and (iii) tail classifier-free guidance that concentrates guidance on the final refinement stage. Figure 1 illustrates the complete pipeline.

3.1 Overview of the Proposed Framework

The pipeline begins by extracting the foreground object from the reference image and composing it with the target background in latent space. The resulting composite latent is projected onto the FLUX editing trajectory using RF-Inversion. During denoising, EEdit’s SLoC mechanism focuses computation on the insertion region while reusing cached representations for distant background tokens.

Our primary modification occurs inside the transformer attention layers. **A separate forward pass over the reference image extracts appearance representations, which are converted into key-value memories and injected into selected FLUX blocks.** This allows image tokens to access reference-specific appearance information throughout generation.

To further improve visual consistency, we augment prompts with domain-aware descriptors and apply classifier-free guidance only during the final denoising iterations. Finally, background anchoring constrains modifications to the insertion region and preserves scene structure outside the edit mask.

3.2 EEdit Backbone

We encode the target background and masked reference object into FLUX latent space using the VAE encoder $\mathcal{E}$, obtaining latents z_b and z_r . The foreground latent is resized and inserted into the target location to obtain an initial composite latent

$$z_c = (1 - M) \odot z_b + M \odot z_r, \quad (1)$$

where M denotes the latent-space insertion mask.

This composition provides a strong object prior before editing begins, allowing subsequent denoising stages to focus primarily on appearance harmonization and scene integration rather than object synthesis from scratch.

Following EEdit, the composite latent is projected onto the FLUX rectified-flow trajectory using RF-Inversion. Let Y_{t_i} denote the latent state at inversion timestep t_i , and let $y_1 \sim \mathcal{N}(0, I)$ be a fixed Gaussian noise sample. The guided inversion velocity is computed as

$$\hat{u}_{t_i} = u_{t_i} + \gamma \left(\frac{y_1 - Y_{t_i}}{1 - t_i} - u_{t_i} \right), \quad (2)$$

where $u_{t_i} = f_\theta(Y_{t_i}, t_i, \emptyset)$ is the unconditional model velocity. The fraction $\frac{y_1 - Y_{t_i}}{1 - t_i}$ is the analytical conditional velocity toward the fixed noise sample y_1 , and γ controls the strength of this trajectory guidance.

We inherit EEdit’s inversion step skipping, recomputing transformer velocities only at fixed intervals and reusing the most recent velocity estimate at intermediate timesteps. We additionally adopt EEdit’s Spatial Locality Caching (SLoC), which exploits the observation that composition modifies only a localized subset of image tokens. Let $\mathcal{E}_0$ denote the insertion region. Each token is assigned an importance score:

$$s_n = 1 + \text{edit_base} \sum_{k=0}^{K-1} \text{bonus_ratio}^{1+k} \mathbf{1}[n \in \mathcal{E}_k \setminus \mathcal{E}_{k-1}], \quad (3)$$

where tokens closer to the insertion region receive higher priority. At each denoising step, only the highest-scoring fraction of tokens is recomputed, while remaining tokens reuse cached hidden states. Together, inversion step skipping and SLoC form the efficient editing backbone inherited from EEdit.

3.3 Reference Attention Injection

Our primary contribution is a reference-guided attention mechanism that explicitly injects appearance information from the reference image into the diffusion transformer.

Although latent composition initializes the object appearance, repeated denoising can progressively weaken this signal. In practice, text prompts alone are often insufficient to preserve fine-grained

visual characteristics such as texture, color distribution, material properties, and stylistic details. Consequently, the generated object may drift away from its reference appearance despite being correctly positioned within the target scene.

To mitigate this issue, we treat the reference image as an external visual memory. Prior to denoising, a separate forward pass is performed on the reference image and intermediate hidden states are extracted from the first B joint transformer blocks. These layers primarily encode semantic appearance and object identity before later layers become increasingly specialized for scene-specific rendering.

For each selected block i , the extracted hidden state h_i^{ref} is projected into attention keys and values:

$$K_i^{\text{ref}} = \text{to}_k(h_i^{\text{ref}}), \quad V_i^{\text{ref}} = \text{to}_v(h_i^{\text{ref}}). \quad (4)$$

The resulting tensors are computed once and cached throughout generation.

During denoising, FLUX constructs standard query, key, and value tensors from image and text tokens. We extend the attention memory by appending the reference keys and values:

$$K_{\text{ext}} = [K; K_i^{\text{ref}}], \quad V_{\text{ext}} = [V; V_i^{\text{ref}}]. \quad (5)$$

Attention is then computed as

$$A = \text{Softmax} \left(\frac{Q K_{\text{ext}}^\top}{\sqrt{d_k}} \right) V_{\text{ext}}. \quad (6)$$

Unlike token concatenation approaches, the reference image participates only as an attention memory source. Reference features contribute keys and values but do not generate output tokens themselves. Consequently, image tokens can retrieve appearance information from the reference object while preserving the spatial layout dictated by the target scene and textual prompt.

We further restrict injection to the first few joint transformer blocks. Early layers primarily govern object identity and semantic appearance, whereas deeper layers increasingly focus on local rendering and texture synthesis. Limiting injection to early layers therefore encourages faithful identity preservation while reducing the risk of copying artifacts or spatial inconsistencies.

3.4 Domain-Aware Composition

Reference attention injection preserves object identity, but successful composition also requires stylistic harmonization with the target scene. This challenge becomes particularly pronounced in cross-domain settings where the same object may need to be integrated into photorealistic, cartoon, sketch, or painted environments.

We observed that EEdit frequently produced inconsistent harmonization across domains because prompt conditioning alone provided insufficient stylistic guidance. To address this limitation, we augment the original prompt with domain-specific descriptors:

$$p_{\text{aug}} = p_{\text{original}} \parallel s_{\text{domain}}, \quad (7)$$

where s_{domain} contains textual descriptors corresponding to the target visual domain.

We additionally introduce tail classifier-free guidance (CFG). Standard CFG applies guidance throughout the entire denoising trajectory, approximately doubling transformer evaluations. However, style adaptation and appearance refinement are most critical during the final denoising stages. We therefore activate CFG only during the final refinement window:

$$\hat{v}_t = \begin{cases} v_\theta(z_t, t, c_{\text{pos}}), & t < T_{\text{start}}, \\ v_\theta(z_t, t, c_{\text{neg}}) + s [v_\theta(z_t, t, c_{\text{pos}}) - v_\theta(z_t, t, c_{\text{neg}})], & t \geq T_{\text{start}}. \end{cases} \quad (8)$$

Finally, background consistency is maintained through latent anchoring. After each denoising step, non-edited regions are restored using the corresponding noised background latent:

$$z_{t-1} \leftarrow (1 - M) \odot z_{\text{bg}}^{(\sigma_t)} + M \odot z_{t-1}. \quad (9)$$

This constrains modifications to the insertion region while preserving scene structure elsewhere. The final latent is decoded using the FLUX VAE decoder

$$\hat{I} = \mathcal{D}(z_0). \quad (10)$$

Together, reference attention injection, domain-aware conditioning, and tail CFG extend the EEdit framework for instance-level image composition, enabling improved object fidelity while retaining the efficiency advantages of the original editing backbone.

4 Experimental Design

Our experiments are designed to evaluate whether the proposed inference-time extensions change the behavior of the EEdit composition pipeline. We focus on three research questions:

RQ1: Does reference K/V attention injection improve semantic alignment with the desired composition relative to the unmodified EEdit pipeline?

RQ2: How does the proposed method affect background preservation across different target visual domains?

RQ3: Do the effects of reference injection and domain-aware guidance vary across cross-domain settings such as Real→Cartoon, Real→Painting, Real→Sketch, and Real→Real?

These questions reflect the central goal of the project: not to replace EEdit as a general-purpose editing framework, but to examine whether explicit reference attention and domain-aware guidance provide a useful alternative operating point for reference-guided image composition.

4.1 Dataset and Evaluation Setup

We evaluate on the **TF-ICON Test Benchmark** (Lu et al., 2023), which is designed for training-free cross-domain image-guided composition. The benchmark provides foreground objects, foreground masks, and target background scenes, making it well aligned with our task of inserting a specific reference object into a target visual domain while preserving object identity and scene consistency.

Unlike PIE-Bench (UB-CVML-Group, 2024), which primarily evaluates prompt-driven edits to a source image, TF-ICON directly evaluates reference-guided composition. Its samples cover heterogeneous target domains, including photorealistic, cartoon, sketch, and painting-style scenes. This allows us to evaluate both same-domain and cross-domain composition behavior.

For each sample, we use the provided foreground object and mask as the reference input, the target scene as the background image, and the associated text prompt as conditioning. The foreground mask defines the object region used for latent composition and reference feature extraction, while the target insertion region defines the latent-space mask used during denoising and background anchoring. Since the method is training-free, no training or validation split is used; all experiments are conducted directly on the benchmark test set.

4.2 Baseline Comparison

We compare our method against the unmodified **EEdit** composition pipeline (Yan et al., 2025). This is the most direct and controlled baseline because our method uses EEdit as its computational backbone and modifies only the denoising process.

Both EEdit and our method use the same foreground object, foreground mask, target background image, prompt, RF-Inversion procedure, inversion step skipping, Spatial Locality Caching, Token Index Preprocessing, and background anchoring. The difference is that our method additionally injects precomputed reference K/V memories into early FLUX attention blocks, augments the prompt with domain-specific descriptors, and applies classifier-free guidance only during the final denoising steps. This controlled comparison isolates the effect of our proposed extensions while keeping the underlying editing framework including any hyperparameters fixed.

Table 1: Quantitative comparison between EEdit and our method on the TF-ICON test benchmark. We report PSNR, SSIM, LPIPS, and CLIP similarity across four cross-domain composition categories. Best results within each category are shown in **bold**.

Category	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	CLIP $\uparrow$
Real $\rightarrow$ Cartoon	EEdit	36.31	0.9767	0.0073	28.42
	Ours	31.37	0.9628	0.0285	28.47
Real $\rightarrow$ Painting	EEdit	33.14	0.9368	0.0194	26.90
	Ours	29.21	0.9177	0.0423	27.14
Real $\rightarrow$ Sketch	EEdit	33.51	0.9531	0.0161	29.72
	Ours	34.43	0.9544	0.0132	30.18
Real $\rightarrow$ Real	EEdit	32.92	0.9164	0.0218	25.35
	Ours	32.97	0.9164	0.0213	25.52
Average	EEdit	33.97	0.9458	0.0162	27.60
	Ours	32.00	0.9361	0.0263	27.83

4.3 Evaluation Metrics

We report four metrics: PSNR, SSIM, LPIPS, and CLIP similarity.

PSNR and **SSIM** measure low-level reconstruction and structural similarity. In our setting, they primarily reflect how well the method preserves the target scene and avoids unnecessary changes outside the intended edit region. Higher values indicate stronger preservation.

LPIPS measures perceptual distance using deep visual features. It complements PSNR and SSIM by capturing perceptual differences that may not be reflected by pixel-level similarity. Lower LPIPS indicates closer perceptual similarity to the reference target.

CLIP similarity measures semantic alignment between the generated composition and the target conditioning. Since our modifications are intended to improve reference-guided and domain-aware composition, CLIP provides a useful proxy for whether the output better matches the intended visual concept. Higher CLIP similarity indicates stronger semantic alignment.

Together, these metrics capture the main trade-off studied in this work: background preservation versus reference-guided semantic alignment.

4.4 Analysis Plan

We analyze results along two axes. First, we compare EEdit and our method within each TF-ICON domain to determine whether the proposed modifications improve CLIP alignment, background preservation, or both. Second, we compare trends across domains to identify where reference injection is beneficial and where it introduces additional background disturbance.

We interpret improvements in CLIP similarity as evidence of stronger semantic or reference-guided alignment, while changes in PSNR, SSIM, and LPIPS indicate the degree to which the method preserves or perturbs the target scene. This analysis allows us to characterize the proposed method as a controllable extension of EEdit rather than as a universally superior replacement.

5 Results

We evaluate our method against the unmodified EEdit composition pipeline on the TF-ICON test benchmark. Results are reported across four cross-domain composition settings: Real $\rightarrow$ Cartoon, Real $\rightarrow$ Painting, Real $\rightarrow$ Sketch, and Real $\rightarrow$ Real. Table 1 reports PSNR, SSIM, LPIPS, and CLIP similarity for both methods.

Across all categories, our method improves CLIP similarity relative to EEdit, increasing the average CLIP score from 27.60 to 27.83. EEdit obtains higher average PSNR, SSIM, and lower LPIPS, with average PSNR decreasing from 33.97 to 32.00, SSIM from 0.9458 to 0.9361, and LPIPS increasing from 0.0162 to 0.0263 for our method.

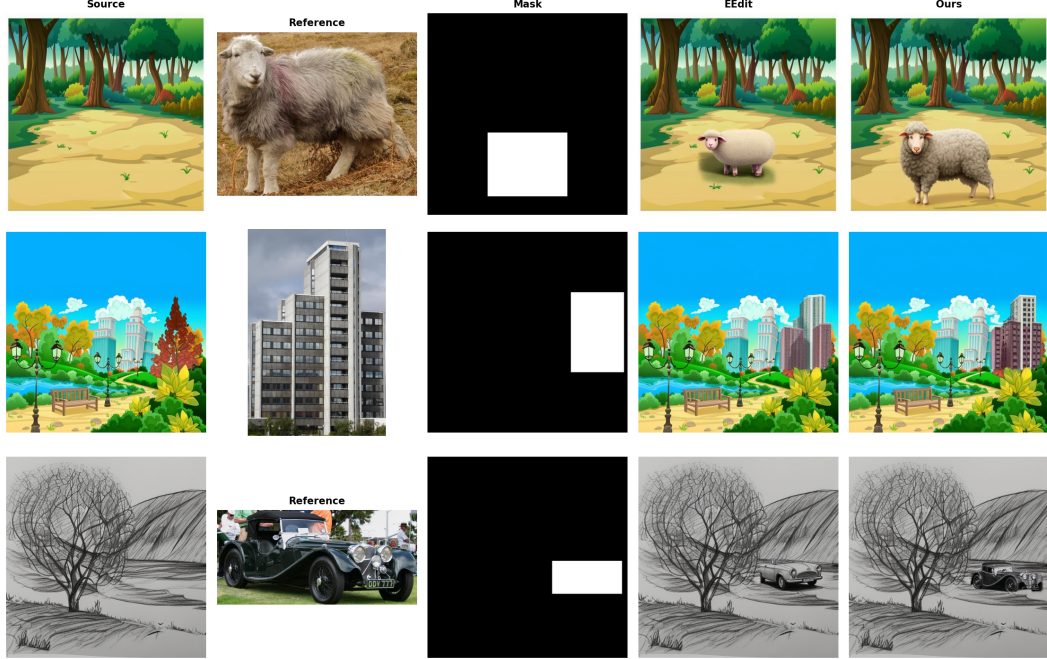


Figure 2: **Qualitative comparison with EEdit** on Real→Cartoon and Real→Sketch examples. Each row shows the source image, reference object, mask, EEdit output, and our output. EEdit provides a strong state-of-the-art composition backbone with stable scene preservation, while our augmentation encourages the inserted object to more closely follow the *reference appearance and target-domain style*, such as the sheep texture, building facade, and car silhouette.

For Real→Sketch, our method improves all four metrics: PSNR increases from 33.51 to 34.43, SSIM from 0.9531 to 0.9544, LPIPS decreases from 0.0161 to 0.0132, and CLIP increases from 29.72 to 30.18. For Real→Real, our method improves PSNR from 32.92 to 32.97, LPIPS from 0.0218 to 0.0213, and CLIP from 25.35 to 25.52, while matching EEdit in SSIM. For Real→Cartoon and Real→Painting, our method improves CLIP similarity but obtains lower background-preservation metrics than EEdit. In Real→Cartoon, CLIP increases from 28.42 to 28.47, while PSNR decreases from 36.31 to 31.37 and LPIPS increases from 0.0073 to 0.0285. In Real→Painting, CLIP increases from 26.90 to 27.14, while PSNR decreases from 33.14 to 29.21 and LPIPS increases from 0.0194 to 0.0423. Figure 2 shows qualitative examples comparing our results with EEdit’s.

6 Discussion

The comparison with EEdit highlights a clear trade-off between strict background preservation and stronger reference-guided composition. Since our method uses EEdit as its backbone, the results in Table 1 should not be interpreted as a broad benchmark against unrelated editing methods. Instead, they isolate the effect of adding reference K/V attention injection, domain-aware prompt augmentation, and tail classifier-free guidance to the EEdit denoising process.

Across all TF-ICON composition categories, our method improves CLIP similarity relative to EEdit. This supports the motivation for reference K/V attention injection: instead of relying only on latent initialization and prompt conditioning, the denoising process receives direct access to reference-derived appearance memories.

The most favorable settings are Real→Sketch and Real→Real, where our method improves or matches EEdit across nearly all metrics. This suggests that reference attention injection is most effective when the target domain either preserves strong structural cues or remains visually close to the reference domain. In contrast, Real→Cartoon and Real→Painting reveal the limitations of the proposed modifications. Although our method slightly improves CLIP similarity in both settings, it reduces PSNR and SSIM and increases LPIPS relative to EEdit. This suggests that stronger

reference-driven denoising and tail CFG can introduce additional visual changes near the insertion region, particularly in domains that require substantial stylization. EEdit remains more conservative in these cases and better preserves low-level background similarity. Our method, by contrast, favors reference/style alignment over strict reconstruction.

These results show that the proposed extensions do not uniformly outperform EEdit, nor are they intended to replace it as a general-purpose acceleration framework. Instead, they explore a different operating point: injecting reference appearance and domain-aware guidance into an already efficient EEdit backbone. The results suggest that this direction is promising for settings where foreground identity and semantic alignment are more important than pixel-level reconstruction, especially in sketch and real-domain composition. However, the degradation in background preservation for cartoon and painting domains indicates that reference injection and tail CFG require more careful control in highly stylized settings.

A natural next step is to make the guidance strength adaptive. For example, the number of injected joint blocks, the duration of reference injection, and the tail-CFG scale could be selected per domain or per image based on the desired trade-off between object fidelity and background preservation. More localized reference injection, restricted to foreground tokens rather than all image tokens, may also reduce background disturbance while retaining the benefits of reference appearance transfer. Overall, the comparison with EEdit suggests that our modifications are most useful as controllable extensions to the EEdit pipeline rather than as a universally superior replacement.

7 Conclusion

In this work, we investigated a lightweight extension of EEdit for instance-level reference-guided image composition. Rather than proposing a new editing backbone, we retained EEdit’s efficient RF-Inversion, inversion step skipping, Spatial Locality Caching, and background anchoring, and modified the denoising process with reference K/V attention injection, domain-aware prompt augmentation, and tail classifier-free guidance. This design allows the FLUX denoising process to access reference-object appearance cues directly through attention memory, instead of relying only on latent initialization and prompt conditioning.

Our experiments on the TF-ICON test benchmark show that these additions shift the behavior of EEdit toward stronger semantic and reference-guided alignment. Compared with the unmodified EEdit pipeline, our method improves CLIP similarity across all evaluated domains, with the clearest gains in Real→Sketch and Real→Real composition. At the same time, results in Real→Cartoon and Real→Painting reveal a trade-off: stronger reference-driven denoising can reduce strict pixel-level background preservation in highly stylized domains. These findings suggest that reference attention injection is a useful controllable extension to EEdit, but not a universal replacement for its conservative reconstruction behavior.

Future work should explore adaptive control of the reference-injection strength, including the number of injected transformer blocks, the duration of injection, and the tail-CFG scale. More localized foreground-only reference injection may also help preserve the benefits of object identity transfer while reducing unintended background disturbance. Overall, our results indicate that exposing the denoising process to explicit reference appearance memories is a promising direction for improving instance-level composition in diffusion transformer editing pipelines.

References

- Black Forest Labs. FLUX.1: A family of flow matching text-to-image models. <https://blackforestlabs.ai>, 2024.
- Tim Brooks, Aleksander Holynski, and Alexei A Efros. Instructpix2pix: Learning to follow image editing instructions. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 18392–18402, 2023.
- Mingdeng Cao, Xintao Wang, Zhongang Qi, Ying Shan, Xiaohu Qie, and Yinqiang Zheng. Masactrl: Tuning-free mutual self-attention control for consistent image synthesis and editing. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 22560–22570, 2023.

- Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross attention control. *arXiv preprint arXiv:2208.01626*, 2022.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.
- Tasin Islam, Alina Miron, Xiaohui Liu, and Yongmin Li. Deep learning in virtual try-on: A comprehensive survey. *IEEE Access*, 12:29475–29502, 2024.
- Bahjat Kawar, Shiran Zada, Oran Lang, Omer Tov, Huiwen Chang, Tali Dekel, Inbar Mosseri, and Michal Irani. Imagic: Text-based real image editing with diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 6007–6017, 2023.
- Shilin Lu, Yanzhu Liu, and Adams Wai-Kin Kong. Tf-icon: Diffusion-based training-free cross-domain image composition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2294–2305, 2023.
- Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. Sdedit: Guided image synthesis and editing with stochastic differential equations. *arXiv preprint arXiv:2108.01073*, 2021.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.
- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*, 2020.
- Narek Tumanyan, Michal Geyer, Shai Bagon, and Tali Dekel. Plug-and-play diffusion features for text-driven image-to-image translation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1921–1930, 2023.
- UB-CVML-Group. PIE-Bench++: A benchmark dataset for prompt-based image editing evaluation. Hugging Face Datasets, https://huggingface.co/datasets/UB-CVML-Group/PIE_Bench_pp, 2024. License: CC BY-SA 4.0.
- Zexuan Yan, Yue Ma, Chang Zou, Wenteng Chen, Qifeng Chen, and Linfeng Zhang. Eedit: Rethinking the spatial and temporal redundancy for efficient image editing. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 17474–17484, 2025.
- Tianrui Zhu, Shiyi Zhang, Jiawei Shao, and Yansong Tang. Kv-edit: Training-free image editing for precise background preservation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16607–16617, 2025.